ACTIVITY1

Done By:

Prajiith K
192324013
B.Tech AI&DS

Data Wrangling in Python

Code:

```
import pandas as pd
import numpy as np
students = {
    "ID": [101, 102, 103, 104, 105, 105],
    "Name": ["Amit", "Priya", "Rahul", "Sneha", "Riya", "Riya"],
    "Age": [20, 21, 20, 22, 21, 21],
    "Gender": ["M", "F", "M", "F", "F", "F"],
    "Marks": [85, np.nan, 78, 92, np.nan, np.nan]
}
fees = {
    "ID": [101, 102, 103, 104, 105],
    "Pending": [0, 500, 1000, 0, 250]
}
employees1 = {
    "Name": ["John", "Alice"],
    "Age": [25, 30],
    "City": ["Chennai", "Delhi"]
}
employees2 = {
    "Name": ["David", "Sara"],
    "Age": [28, 27],
    "City": ["Mumbai", "Pune"]
}

cars = {
    "Brand": ["Maruti", "Maruti", "Hyundai", "Toyota", "Hyundai", "Toyota"],
    "Year": [2020, 2021, 2020, 2021, 2020, 2021],
    "Sold": [5, 8, 6, 4, 7, 3]
}
```

```
df = pd.DataFrame(students)
print("ORIGINAL DATAFRAME")
print(df)


print("\nMISSING VALUES")
print(df.isnull().sum())
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
print("\nAFTER FILLING MISSING VALUES")
print(df)
```

```
AFTER FILLING MISSING VALUES
    ID   Name  Age Gender  Marks
0  101   Amit   20      M   85.0
1  102  Priya   21      F   85.0
2  103  Rahul   20      M   78.0
3  104  Sneha   22      F   92.0
4  105   Riya   21      F   85.0
5  105   Riya   21      F   85.0
```

```
df["Gender"] = df["Gender"].replace({"M": 0, "F": 1})
print("\nAFTER REPLACING GENDER")
print(df)
```

```
AFTER REPLACING GENDER
    ID   Name  Age  Gender  Marks
0  101   Amit   20       0   85.0
1  102  Priya   21       1   85.0
2  103  Rahul   20       0   78.0
3  104  Sneha   22       1   92.0
4  105   Riya   21       1   85.0
5  105   Riya   21       1   85.0
```

```
filtered = df[df["Marks"] >= 80]
print("\nFILTERED DATA (Marks >= 80)")
print(filtered)
```

```
FILTERED DATA (Marks >= 80)
    ID   Name  Age  Gender  Marks
0  101   Amit   20       0   85.0
1  102  Priya   21       1   85.0
3  104  Sneha   22       1   92.0
4  105   Riya   21       1   85.0
5  105   Riya   21       1   85.0
```

```
details = df[["ID", "Name", "Age", "Gender", "Marks"]]

fees_df = pd.DataFrame(fees)

merged = pd.merge(details, fees_df, on="ID")

print("\nMERGED DATA")

print(merged)
```

```
MERGED DATA
    ID   Name  Age  Gender  Marks  Pending
0  101   Amit   20       0   85.0        0
1  102  Priya   21       1   85.0      500
2  103  Rahul   20       0   78.0     1000
3  104  Sneha   22       1   92.0        0
4  105   Riya   21       1   85.0      250
5  105   Riya   21       1   85.0      250
```

```
car_df = pd.DataFrame(cars)

print("\nCAR DATA")

print("\nGROUP BY YEAR")

print(car_df.groupby("Year").sum(numeric_only=True))
```

```
GROUP BY YEAR
      Sold
Year
2020    18
2021    15
```

```
print("\nGROUP BY BRAND")

print(car_df.groupby("Brand")["Sold"].sum())
```

```
GROUP BY BRAND
Brand
Hyundai    13
Maruti     13
Toyota      7
```

```
print("\nDATA FOR YEAR 2020")

print(car_df.groupby("Year").get_group(2020))
```

```
DATA FOR YEAR 2020
     Brand  Year  Sold
0   Maruti  2020     5
2  Hyundai  2020     6
4  Hyundai  2020     7
```

```
print("\nDUPLICATE ROWS")
print(df[df.duplicated()])

no_duplicate = df.drop_duplicates()

print("\nAFTER REMOVING DUPLICATES")
print(no_duplicate)
```

```
AFTER REMOVING DUPLICATES
    ID   Name  Age  Gender  Marks
0  101   Amit   20       0   85.0
1  102  Priya   21       1   85.0
2  103  Rahul   20       0   78.0
3  104  Sneha   22       1   92.0
4  105   Riya   21       1   85.0
```

```
emp1 = pd.DataFrame(employees1)
emp2 = pd.DataFrame(employees2)

concat_df = pd.concat([emp1, emp2], ignore_index=True)

print("\nCONCATENATED DATAFRAME")
print(concat_df)
```

```
CONCATENATED DATAFRAME
    Name  Age     City
0   John   25  Chennai
1  Alice   30    Delhi
2  David   28   Mumbai
3   Sara   27     Pune
```